# Cervical Human Vertebrae
## Quantitative Three-Dimensional Anatomy of the Middle and Lower Regions

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**In this study, the three-dimensional quantitative anatomy of middle and lower cervical vertebrae was determined. The three-dimensional coordinates of various marked points on the surface of the vertebra were measured with a specially designed morphometer instrument. From these coordinates, linear dimensions, angulations, and areas of surfaces and cross-sections of most vertebral components were calculated. The results showed two distinct transition regions: 1) toward the thoracic spine by the wider C7 vertebra but narrower spinal canal; and 2) toward the upper cervical region with the larger pedicle and spinous process of C2. Based on the study of 72 human cervical vertebrae, mean and standard error of the mean values of some clinically important dimensions of vertebral body, spinal canal, pedicles, transverse processes, spinous process, and uncovertebral joints are given for C2–C7 vertebrae. The areas of the end plates, spinal canal, and pedicles were modeled by elliptical and triangular shapes, and results were compared with the actual measurements. [Key words: cervical spine, vertebrae, anatomy, vertebral dimensions, spinal canal, pedicles]**

THE ANATOMY OF THE CERVICAL SPINE has been documented qualitatively in various textbooks.[6,7] However, accurate quantitative anatomic data rarely are reported. Some quantitative studies of human cervical vertebral anatomy have been completed, but most of these studies use relatively simplistic measurement techniques[5,10] and/or present anatomic data on a small portion of the total vertebral anatomy.[9] Quantitative anatomic studies of the thoracolumbar vertebrae are reported more frequently[2,8,14,15] because of the need for accurate size and orientation information of the pedicles.

The importance of quantitative anatomy lies primarily in the development of mathematical models of the spine, although it also has considerable use clinically in the diagnosis of disease or injury and in designing surgical implants and procedures. To better understand injury mechanisms, spinal kinematics, and instability, various three-dimensional kinetic/kinematic models have been developed.[1,3,11] Proper implementation of these models requires accurate three-dimensional quantitative anatomy.

Francis[5] performed a large study using 284 cervical vertebrae (male and female, whites and blacks, aged 25–36 years). Using a caliper, he measured the lateral widths and anteroposterior depths of the vertebral foramens and vertebral bodies, and largest diameters of facets in planes parallel to sagittal and frontal planes. Nissan and Gilad[10] performed a two-dimensional quantitative study of cervical vertebrae. Using calipers, they measured the lateral radiographic projections of cervical spines of 157 subjects and documented the sagittal plane dimensions of various vertebral parts. More recently, Liu et al[9] reported some geometric data of the cervical vertebrae of six cadavers, using digitized points on the vertebral surface.

The purpose of the present study was to quantify the anatomy of several clinically important vertebral parts of the middle and lower cervical regions in three dimensions. The variations in the anatomic dimensions along the length of the spine were documented. In addition, the morphology of some vertebral components was approximated by simple geometric shapes, and results were compared with actual measurements.

## METHODS

The data presented here represent the second of a series of anatomic studies; a detailed description of the methods is found in the initial study, which is published in this issue (p. 888).[12] A brief description follows.

Twelve fresh autopsy spine specimens were obtained and radiographed to identify any significant abnormalities. The specimens were identified by age, sex, and height. The subjects had an average age of 46.3 years (range, 19–59 years), weight of 67.8 kg (range, 54–85 kg), height of 167.8 cm (range, 157–178 cm), and male-to-female ratio of 8 to 4. Individual details are given in the initial study.[12] Each specimen was prepared by complete removal of all soft tissues from the vertebrae. Two pointed threaded rods (2-mm diameter and 2.5-mm long) were secured into each vertebral body. The two ends of one rod constituted two pin markers; the third pin marker was provided by the end of the second threaded rod. Anteroposterior and lateral radiographs were taken to identify the pins in the vertebral body coordinate system with origin at the center of the upper end plate (Figure 1).

Linear dimensions, angulations, and areas of various vertebral parts were studied. A summary of all measured dimensions is listed in Table 1 and shown graphically in Figure 1. Each vertebral part was defined in space by a series of points spanning or circumventing its surface. For example, each uncovertebral facet was defined by points—up to 20 points—spanning its entire surface, whereas each end plate, spinous process, transverse process, and pedicle was defined by points circumventing the structure. The number of points chosen was such that the geometry was well defined by straight lines joining the adjacent points.

The three-dimensional coordinates of all the points with respect to a reference system defined by the three pin markers were determined using a specially designed morphometer.[12,13] This instrument consisted of one linear variable (LVDT) and two rotational variable (RVDT) displacement transducers arranged such that their axes meet at one point, thus establishing a spherical coordinate system (with coordinates $r, \theta, \phi$). The location of any point, or series of points, with respect to the three pin markers, was determined by placing the tip of the morphometer at the point(s) in question and at the three pin markers. The three-dimensional coordinates were directly recorded into the com-

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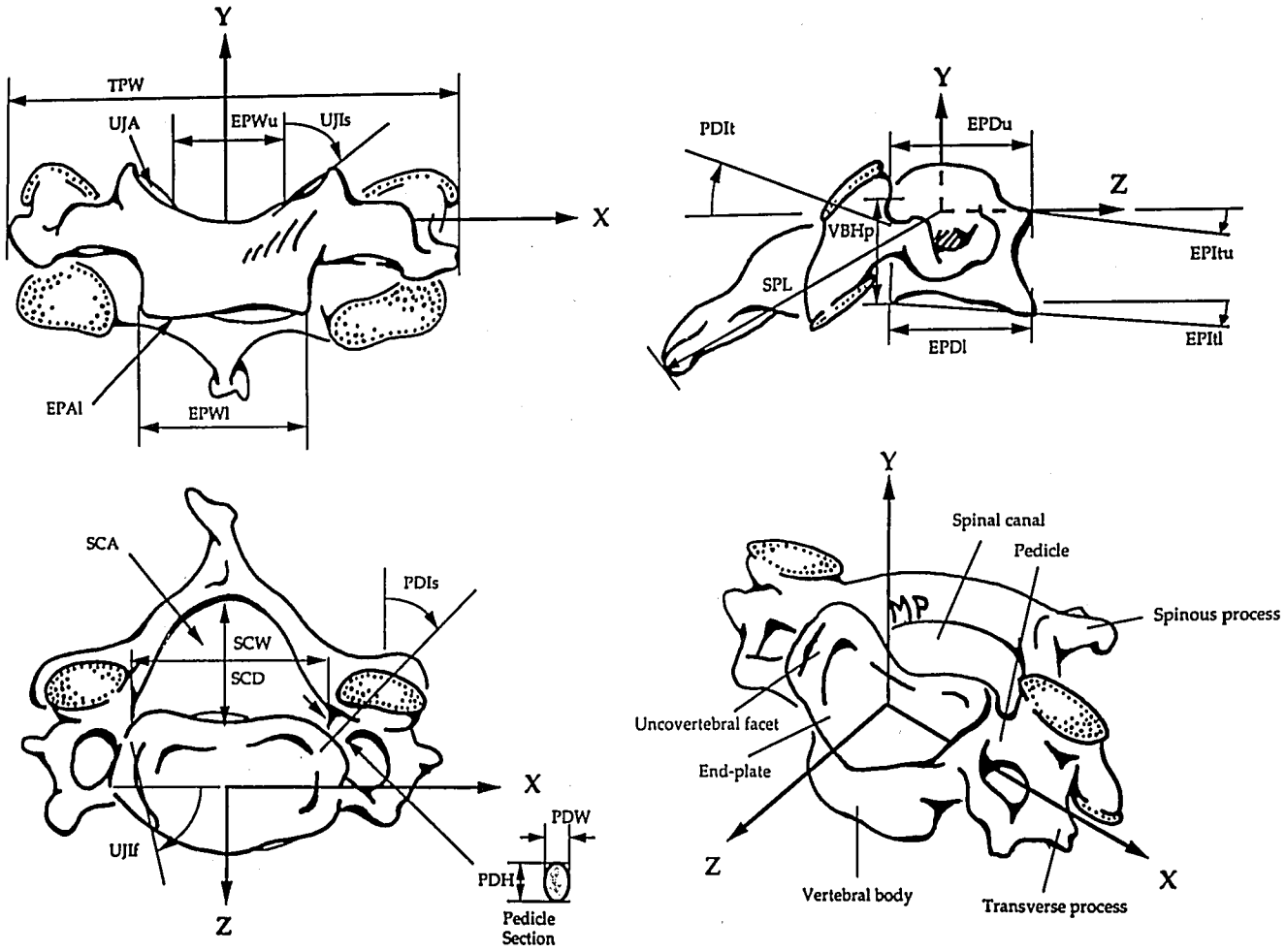



**Fig 1.** Three orthogonal views (front, side, and top) of a cervical vertebra are shown. The various dimensions determined in the present study are indicated and further explained in the nomenclature (Table 1). The fourth view is an isometric view showing the coordinate system used.

puter. In this manner, the morphometer was used to record the three-dimensional positions of all points that defined the vertebral parts listed in Table 1. In addition to the parts listed in Table 1, four points defining the left and right lower and upper edges of the posterior wall of the vertebral body were also digitized. These four points were used to define the vertebral body reference planes: frontal (xy), transverse (xz), and sagittal (yz), using a least-squares technique. The position of the center of the upper end-plate was digitized as the origin for the vertebral body coordinate system (Figure 1).

As stated earlier, three types of anatomic parameters were computed and are described below.

**Linear Parameters.** Computer software was developed to calculate the various linear parameters from the three-dimensional digitized coordinates. The major and minor linear dimensions of each vertebral part were calculated. For example, each pedicle was described by its width and height. In addition, various gross vertebral body dimensions were calculated, such as the height of the vertebral body and the distances between transverse processes. The results of 12 specimens were pooled to calculate mean and standard error of the mean for each linear dimension.

**Surface and Cross-Sectional Area Parameters.** The projected areas of the digitized parts on to the frontal (xy), sagittal (yz) and transverse (zx) planes were computed by considering only two of the three coordinates of each point constituting the part. For example, to calculate frontal plane area, only the x and y coordinates of the points would be considered. The total area was computed as the Euclidean (ie, square root of the sum of the individual squares) sum of the three projected areas. The means and standard error of the means of all specimens were calculated.

**Angular Parameters.** To calculate angulations, the surface of each vertebral part was represented by a plane. Since each part was characterized by more than three points, the equation of a plane was used to yield the plane that would represent the best fit to the surface, using the least-squares techniques. The angles between the best fit plane and each anatomic plane were obtained.

**Statistical Analysis.** For all parameters, differences between vertebral levels were determined with a single-factor analysis of variance (ANOVA) and the Scheffe *F*-test for pair-wise comparison. Note that intercomparisons between groups using the Student *t*-test when more than two groups are involved is not a valid statistical test.

**Accuracy of Measurements.** The accuracy in computing linear, angular, and surface area parameters was estimated with an object with known values. A calibration piece with sides cut at 30°, 45°, and 60° was manufactured out of Plexiglas. A sheet of millimeter graph paper was pasted on each of the cut surfaces, and different geometric shapes were drawn on it. The three pin markers, similar to those placed on a vertebra, were identified on the Plexiglas piece. Thus a pseudovertebra with surfaces of known inclinations and parts of known dimensions and areas was created. The parts were digitized with the same protocol and

Table 1. Table of Nomenclature (Refer to Figure 1)

| | *Mnemonics* | *Dimension* |
|---|---|---|
| Body Part | | |
| Vertebral body | EPW | End-Plate Width |
| | EPD | End-Plate Depth |
| | EPA | End-Plate Area |
| | EPI | End-Plate Inclination |
| | VBH | Vertebral Body Height |
| Spinal canal | | |
| | SCW | Spinal Canal Width |
| | SCD | Spinal Canal Depth |
| | SCA | Spinal Canal Area |
| Pedicle | | |
| | PDW | Pedicle Width |
| | PDH | Pedicle Height |
| | PDA | Pedicle Area |
| | PDI | Pedicle Inclination |
| Spinous process | | |
| | SPL | Spinous Process Length |
| Transverse process | | |
| | TPW | Transverse Process Width |
| Uncovertebral joint | | |
| | UJI | Uncovertebral Joint Inclination |
| | UJA | Uncovertebral Joint Area |
| Suffixes | | |
| | a | Anterior |
| | p | Posterior |
| | s | Sagittal plane |
| | t | Transverse plane |
| | f | Frontal plane |
| | u | Upper |
| | l | Lower (or left, depending on context) |
| | r | Right |

*Note:* The mnemonics for each body part consists of three upper case letters, the first two representing the part and the third representing the measured dimension. This three-letter combination is followed by lower case characters (suffixes) indicating direction, plane and left/right, etc.

computer programs adopted for the vertebral parts. The linear, angular, and surface-area parameters were computed and compared with the known values. The error was found to be less than 5% in all cases.

## RESULTS

Twelve specimens were studied. Therefore, this article describes the results from 72 middle and lower cervical (C2–C7) vertebrae. The eight male and four female specimens were grouped together. We acknowledge differences in the anatomy between the sexes, as outlined by Francis,[5] but have not focused on them in this study. The means and standard error of the means of all measured parameters are tabulated in Tables 2–6 for the vertebral body, spinal canal, pedicles, spinous processes, transverse processes, and uncovertebral joints. Statistical differences for each anatomic parameter between vertebral levels are included in these tables.

### Vertebral Body

The widths and depths of both upper and lower end-plates increased from C2 through C7 (Figure 2A). For the upper end-plate, a significant increase in both width and depth occurred at C7. For the lower end-plates, the greatest increase in width was at C6, in depth at C5, with a decrease in depth at C7. The end-plate widths were always greater than the corresponding depths. The width-to-depth ratio of the upper end-plate varied little from C3 to C6, with an increase at C7. Similarly, the ratio of the lower end-plate remained fairly constant from C2 to C5, with increases at C6 and C7 (Figure 2B).

The total end-plate area increased from C2 to C7, as shown by thick lines in Figure 2C, with the exception of the lower end-plate from C2 to C3 and from C6 to C7. The areas rose significantly at C7 for the upper end-plate and at C6 for the lower end-plate. In an attempt to validate a hypothesis that the end plates are elliptical, surface areas of the end plates were approximated as ellipses using the measured width and depth ($A = \pi WD/4$). Similar calculations were made assuming the end-plate surfaces to be triangular ($A = WD/2$). These approximated

Table 2. Vertebral Body Dimensions

| | *Cervical Vertebral Level* | | | | | |
|---|---|---|---|---|---|---|
| | *C2* | *C3* | *C4* | *C5* | *C6* | *C7* |
| Linear dimensions (mm) | | | | | | |
| EPDu | — | 15.0 | 15.3 | 15.2 | 16.4 | 18.1 (C3-5) |
| | — | 0.55 | 0.75 | 0.35 | 0.52 | 0.66 |
| EPWu | — | 15.8 | 17.2 | 17.5 | 18.5 (C3) | 21.8 (C3-6) |
| | — | 0.46 | 0.66 | 0.58 | 0.55 | 0.66 |
| EPDl | 15.6 | 15.6 | 15.9 | 17.9 | 18.5 (C2-4) | 16.8 |
| | 0.58 | 0.40 | 0.38 | 0.52 | 0.69 | 0.32 |
| EPWl | 17.5 | 17.2 | 17.0 | 19.4 | 22.0 (C2-4) | 23.4 (C2-5) |
| | 0.52 | 0.29 | 0.49 | 0.40 | 0.75 | 0.98 |
| VBHp | — | 11.6 | 11.4 | 11.4 | 10.9 | 12.8 (C6) |
| | — | 0.35 | 0.43 | 0.32 | 0.26 | 0.46 |
| EPWu/EPDu | — | 1.05 | 1.12 | 1.15 | 1.13 | 1.20 |
| EPWl/EPDl | 1.12 | 1.10 | 1.07 | 1.08 | 1.19 | 1.39 |
| Surface area ($mm^2$) | | | | | | |
| EPAu | — | 169.4 | 183.0 | 182.9 | 221.2 (C3) | 278.3 (C3-6) |
| | — | 8.78 | 9.32 | 7.77 | 7.62 | 12.93 |
| EPAl | 194.4 | 190.7 | 199.2 | 246.2 | 289.9 (C2-4) | 280.3 (C2-4) |
| | 9.64 | 5.74 | 7.27 | 12.07 | 16.54 | 13.71 |
| Angular dimensions (°) | | | | | | |
| EPltu | — | 2.7 | 3.5 | 1.7 | 4.7 | 2.2 |
| | — | 0.77 | 1.5 | 0.43 | 1.1 | 0.54 |
| EPltl | 4.2 | 4.0 | 2.1 | 2.7 | 2.7 | 1.8 |
| | 1.2 | 0.84 | 0.4 | 1.1 | 0.69 | 0.42 |

*Note:* For each parameter, the upper row shows the means and the lower row shows the standard error of the mean. For each parameter, an analysis of variance (ANOVA) was performed to determine which vertebral levels were significantly different ($P < 0.05$), in comparison to the vertebrae shown in parenthesis. For nomenclature, see Table 1 and Figure 1.

Table 3. Spinal Canal Dimensions

| | Cervical Vertebral Level | | | | | |
|---|---|---|---|---|---|---|
| | C2 | C3 | C4 | C5 | C6 | C7 |
| Linear dimensions (mm) | | | | | | |
| SCW | 24.5 | 22.9 | 24.7 | 24.9 | 25.8 | 24.5 |
| | 0.61 | 1.15 | 0.52 | 1.13 | 0.66 | 0.92 |
| SCD | 21.0 (C3,C7) | 16.2 | 17.7 | 17.4 | 18.1 | 15.2 |
| | 0.35 | 1.33 | 0.46 | 0.72 | 0.46 | 1.15 |
| SCW/SCD | 1.17 | 1.41 | 1.40 | 1.43 | 1.42 | 1.61 |
| Cross-sectional areas ($mm^2$) | | | | | | |
| SCA | 374.5 (C7) | 248.7 | 272.0 | 249.5 | 266.5 | 223.8 |
| | 13.08 | 34.21 | 26.18 | 34.09 | 24.94 | 36.26 |

*Note:* For each parameter, the upper row shows the means and the lower row shows the standard error of the mean. For each parameter, ANOVA was performed to determine which vertebral levels were significantly different ($P < 0.05$), in comparison to the vertebrae shown in parenthesis. For nomenclature, see Table 1 and Figure 1.

Table 4. Pedicle Dimensions

| | Cervical Vertebral Level | | | | | |
|---|---|---|---|---|---|---|
| | C2 | C3 | C4 | C5 | C6 | C7 |
| Linear Dimensions (mm) | | | | | | |
| PDWr | 7.7 | 5.8 | 5.7 | 6.1 | 6.3 | 6.6 |
| | 0.35 | 0.64 | 0.60 | 0.46 | 0.49 | 0.42 |
| PDHr | 9.4 (C3-7) | 7.6 | 7.4 | 6.7 | 7.1 | 7.5 |
| | 0.46 | 0.46 | 0.42 | 0.42 | 0.39 | 0.32 |
| PDWl | 8.3 (C3-7) | 5.4 | 5.1 | 5.1 | 5.6 | 6.5 (C4-5) |
| | 0.07 | 0.32 | 0.46 | 0.35 | 0.39 | 0.35 |
| PDHl | 11.1 (C3-7) | 7.2 | 7.3 | 7.3 | 7.5 | 7.5 |
| | 0.14 | 0.35 | 0.49 | 0.39 | 0.32 | 0.28 |
| PDHr/PDWr | 1.22 | 1.3 | 1.3 | 1.1 | 1.1 | 1.1 |
| PDHl/PDWl | 1.34 | 1.3 | 1.43 | 1.43 | 1.34 | 1.15 |
| Cross-sectional areas ($mm^2$) | | | | | | |
| PDAr | 32.3 | 24.2 | 24.7 | 23.8 | 24.5 | 30.4 |
| | 2.44 | 2.69 | 2.65 | 2.83 | 2.58 | 3.08 |
| PDAl | 51.8 (C3-7) | 21.4 | 24.0 | 23.9 | 27.9 | 27.8 |
| | 0.28 | 2.02 | 3.08 | 2.58 | 2.47 | 2.37 |
| Angular Dimensions (°) | | | | | | |
| PDIsr | — | 41.6 | 44.6 | 39.3 | 29.6 (C3-5) | 33.1 (C4) |
| | — | 1.13 | 1.66 | 4.45 | 2.30 | 2.23 |
| PDItr | — | −9.2 | −8.6 | −6.3 | 7.1 (C3-5) | 13.4 (C3-5) |
| | — | 2.58 | 2.44 | 1.63 | 1.13 | 1.70 |
| PDIsl | — | 42.9 | 43.9 | 41.2 | 34.1 | 26.7 (C3-5) |
| | — | 2.16 | 2.47 | 4.77 | 2.16 | 2.69 |
| PDItl | — | −7.1 | −6.5 | −5.7 | 5.9 (C3-5) | 9.2 (C3-5) |
| | — | 0.67 | 1.66 | 1.48 | 2.51 | 1.52 |

*Note:* For each parameter, the upper row shows the means and the lower row shows the standard error of the mean. For each parameter, ANOVA was performed to determine which vertebral levels were significantly different ($P < 0.05$), in comparison to the vertebrae shown in parenthesis. For nomenclature, see Table 1 and Figure 1.

Table 5. Spinous Process and Transverse Process

| | Cervical Vertebral Level | | | | | |
|---|---|---|---|---|---|---|
| | C2 | C3 | C4 | C5 | C6 | C7 |
| Linear Dimensions (mm) | | | | | | |
| SPL | 33.7 | 29.6 | 30.3 | 28.5 | 34.2 | 45.7 (C2-6) |
| | 1.39 | 0.78 | 1.07 | 0.98 | 1.88 | 0.84 |
| TPW | 52.6 | 50.3 | 48.5 | 46.4 | 49.5 | 66.6 (C2-6) |
| | 2.08 | 1.62 | 2.14 | 2.97 | 2.11 | 1.13 |

*Note:* For each parameter, the upper row shows the mean and the lower row shows the standard error of the mean. For each parameter, ANOVA was performed to determine which vertebral levels were significantly different ($P < 0.05$), in comparison to the vertebrae shown in parenthesis. For nomenclature, see Table 1 and Figure 1.

areas, represented by dashed lines, are compared with the measured areas and triangular approximations in Figure 2C. For all levels, the elliptical approximation overestimated the areas by approximately 10%, whereas the triangular approximation was a 40% underestimate.

The upper end-plates were found to be oriented with a tilt of 2–5° from the transverse plane, whereas the lower end-plates exhibited angles of 2–4° from the transverse plane for bodies C3–C7. The angular difference between the upper and lower end-plates was less than 2° at all levels, with no significant trends.

Vertebral body heights on the posterior wall in the midsagittal plane remained relatively constant from C3 through C7, exhibiting a range of 10.9–12.8 mm.

Table 6. Uncovertebral Joints

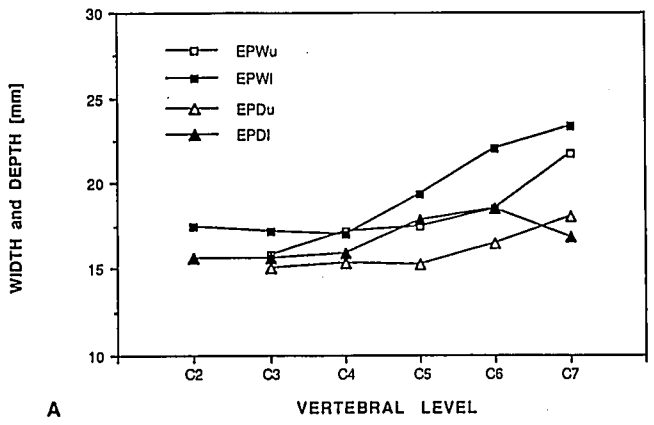
| | Cervical Vertebral Level | | | | | |
|---|---|---|---|---|---|---|
| | C2 | C3 | C4 | C5 | C6 | C7 |
| Surface Areas ($mm^2$) | | | | | | |
| UJAru | — | 46.7 | 40.9 | 45.3 | 48.8 | 43.1 |
| | — | 4.27 | 3.18 | 6.09 | 3.32 | 3.84 |
| UJAlu | — | 40.5 | 37.0 | 40.4 | 58.1 | 41.2 |
| | — | 3.95 | 3.41 | 4.73 | 7.45 | 3.15 |
| UJArl | 19.8 | 22.6 | 25.5 | 27.2 | 25.6 | 23.4 |
| | 3.70 | 2.51 | 3.26 | 2.94 | 2.66 | 2.71 |
| UJAll | 17.7 | 22.3 | 23.5 | 30.5 | 23.9 | 19.0 |
| | 2.40 | 2.89 | 1.96 | 3.84 | 1.79 | 3.00 |
| Angular Dimensions (°) | | | | | | |
| UJIfu | — | 76.6 | 76.2 | 82.7 | 104.0 (C3-5) | 115.6 (C3-6) |
| | — | 2.22 | 2.45 | 1.44 | 2.40 | 2.68 |
| UJIsu | — | 38.7 | 40.0 | 34.5 | 40.8 | 47.3 |
| | — | 3.87 | 2.89 | 2.08 | 3.52 | 3.00 |
| UJIfl | 78.4 | 81.6 | 83.5 | 84.9 | 106.2 (C2-5) | 113.4 (C2-5) |
| | 3.20 | 1.59 | 1.39 | 1.10 | 2.28 | 2.17 |
| UJIsl | 63.7 (C5) | 47.8 | 47.8 | 45.0 | 49.2 | 59.8 |
| | 5.05 | 4.01 | 3.46 | 3.03 | 4.33 | 4.47 |

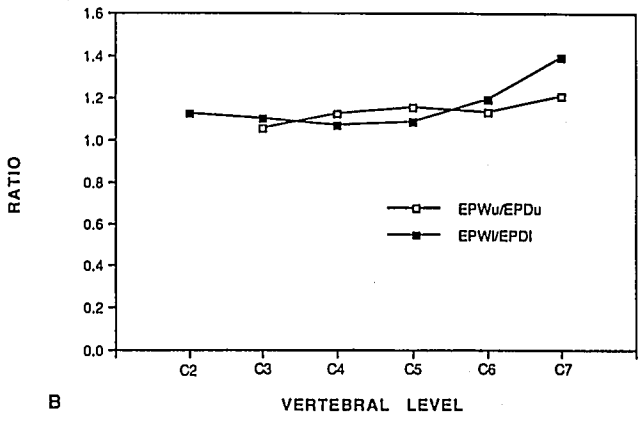
*Note:* For each parameter, the upper row shows the mean and the lower row shows the standard error of the mean. For each parameter, ANOVA was performed to determine which vertebral levels were significantly different ($P < 0.05$), in comparison to the vertebrae shown in parenthesis. For nomenclature, see Table 1 and Figure 1.

## Spinal Canal

The lateral width of the spinal canal was significantly greater than the anteroposterior depth at all levels of the cervical spine (Figure 3A). The width of the canal remained constant from C2 to C7; the depth was greatest at C2 and dropped at C3, where it remained relatively constant to C6 with a slight decrease at C7. As a result, the ratio of lateral width to anteroposterior depth was lowest at C2, constant from C3 to C6, and maximum at C7. These trends are shown in Figure 3B.

The cross-sectional area of the spinal canal at each level was determined and is shown in Figure 3C. The largest area was at C2 and the smallest at C7. There was no significant change between levels C3 through C6. Again, to see if the areas could be approximated by a simple geometric shape, the spinal canal areas were approximated as



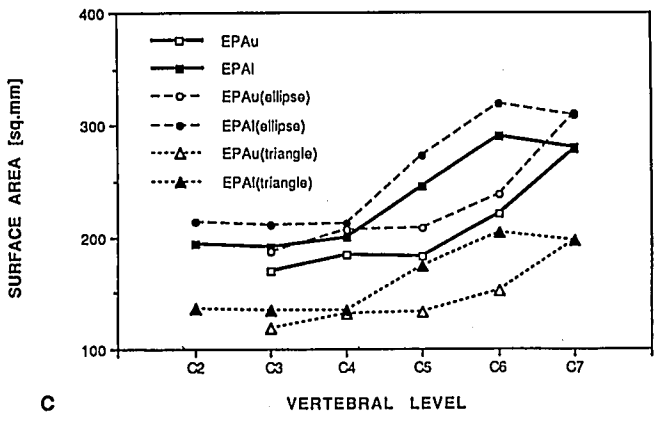




**Fig 2.** Vertebral end plate dimensions as functions of vertebral levels C2–C7 are shown. **A,** Linear dimensions of the upper and lower end-plates. **B,** Ratio of end-plate width to depth for the upper and lower end-plates. **C,** Surface areas of upper and lower-end plates are indicated by whole lines. Dash lines depict the end-plate areas assuming the end-plate shapes to be either an ellipse or a triangle.

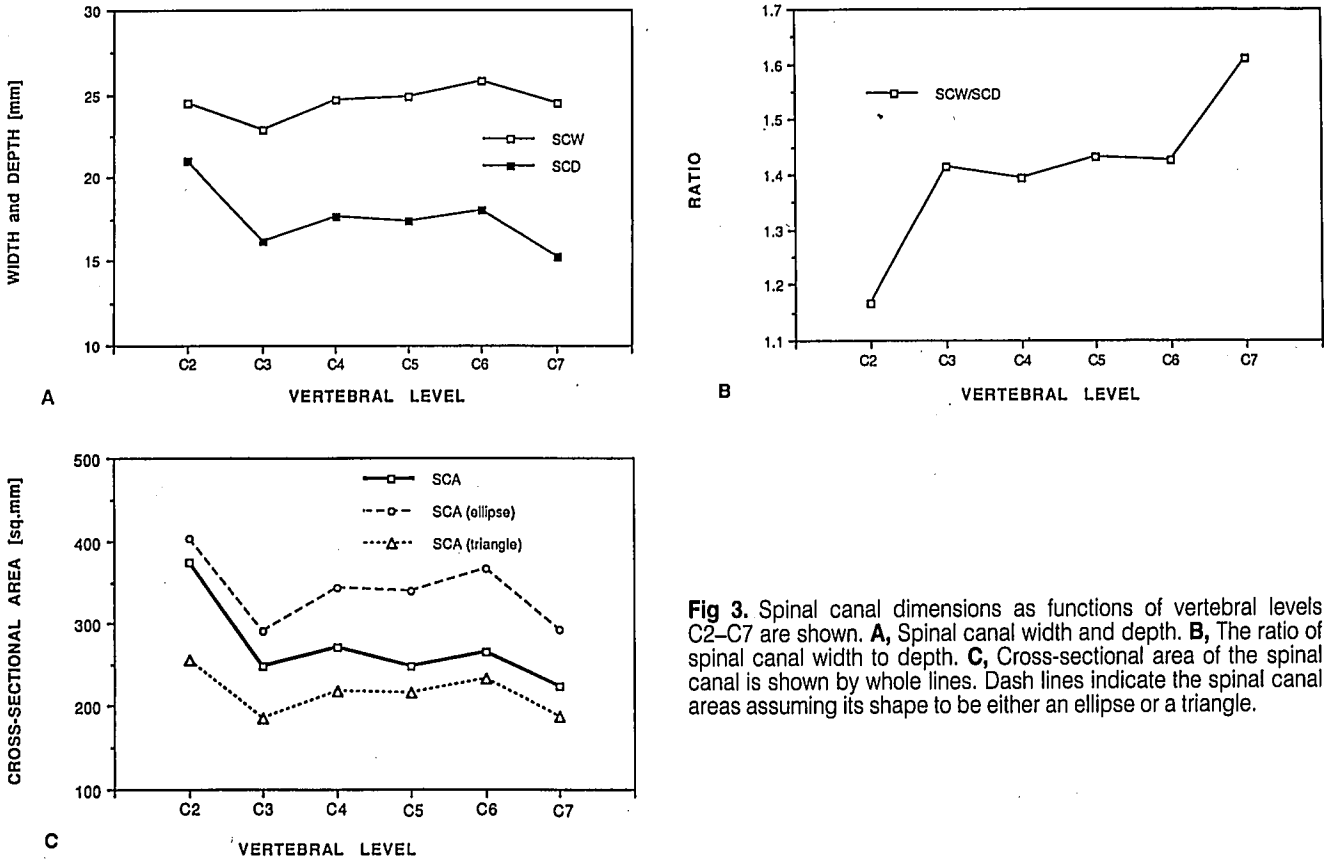



**Fig 3.** Spinal canal dimensions as functions of vertebral levels C2–C7 are shown. **A,** Spinal canal width and depth. **B,** The ratio of spinal canal width to depth. **C,** Cross-sectional area of the spinal canal is shown by whole lines. Dash lines indicate the spinal canal areas assuming its shape to be either an ellipse or a triangle.

ellipses and as triangles and compared with the measured areas in Figure 3C. At the upper levels (C2 and C3), the elliptical approximation was best; at levels C4 through C7, the triangular approximation produced better results.

## Pedicles

The widths and heights of both left and right pedicles are shown in Figure 4A. Both parameters were largest at C2 and smallest at C3, rising slightly to C7. For both left and right pedicles of each vertebra, the pedicle height was significantly greater than its width. These height-to-width ratios are shown in Figure 4B.

The cross-sectional area perpendicular to the pedicle axis for both left and right pedicles was greater at C2 than at all lower levels (Figure 4C). There was large variation at C2, where the total area in our population for the left pedicle was 60% greater than for the corresponding right pedicle. Again, the cross-sectional areas of the pedicles were approximated as ellipses and triangles (Figure 4C). The elliptical approximation was an overestimate and the triangular approximation an underestimate. However, the triangular approximation produced areas closest to the measured values.

From level C3 to C7, the angulation of both left and right pedicles with the transverse plane changed from about 8° below to 11° above the transverse plane. In addition, from C3 to C7, the angle the pedicle axes made with the sagittal plane decreased from about 40° to about 29° (Figure 4D). The direction of the pedicle axes were such that the left and right axes crossed each other in the anterior part of the vertebral body.

## Spinous Process

The three-dimensional length of the spinous process measured from the lower posterior tip to the vertebral body centroid are shown in Figure 5. This length decreased slightly from C2 to C3, remained constant through C5, and significantly increased at C7.

## Transverse Process

The three-dimensional distance between left and right transverse process tips are shown in Figure 5. These distances decreased slightly from C2 to C5 and increased significantly at C6 and C7, showing variation similar to that of the spinous process length.

## Uncovertebral Joint

We measured both the articulating surfaces of the left and right uncovertebral joints. There was no significant trend in the surface areas from C2 to C7 for either the upper or lower articulation surface. The area of the upper articulating surface averaged 44 $mm^2$ and was approximately twice that of the lower articulation.

The uncovertebral joint angulations with respect to the frontal and sagittal planes are shown in Figure 6. For the superior and inferior surfaces, the angle with the frontal plane increased significantly from C5 to C7. The sagittal plane angle remained relatively constant, C2 and C7 having the largest values.

# DISCUSSION

This study presents, for the first time, a comprehensive three-dimensional quantitative data set of the anatomy of the middle and lower cervical vertebrae. Aside from providing a useful database, several interesting points regarding the functional anatomy of the cervical vertebrae were noted and are discussed below.

## Transition Regions

The transition from cervical to the thoracic region was readily apparent from the vertebral body dimensions. The upper end-plate

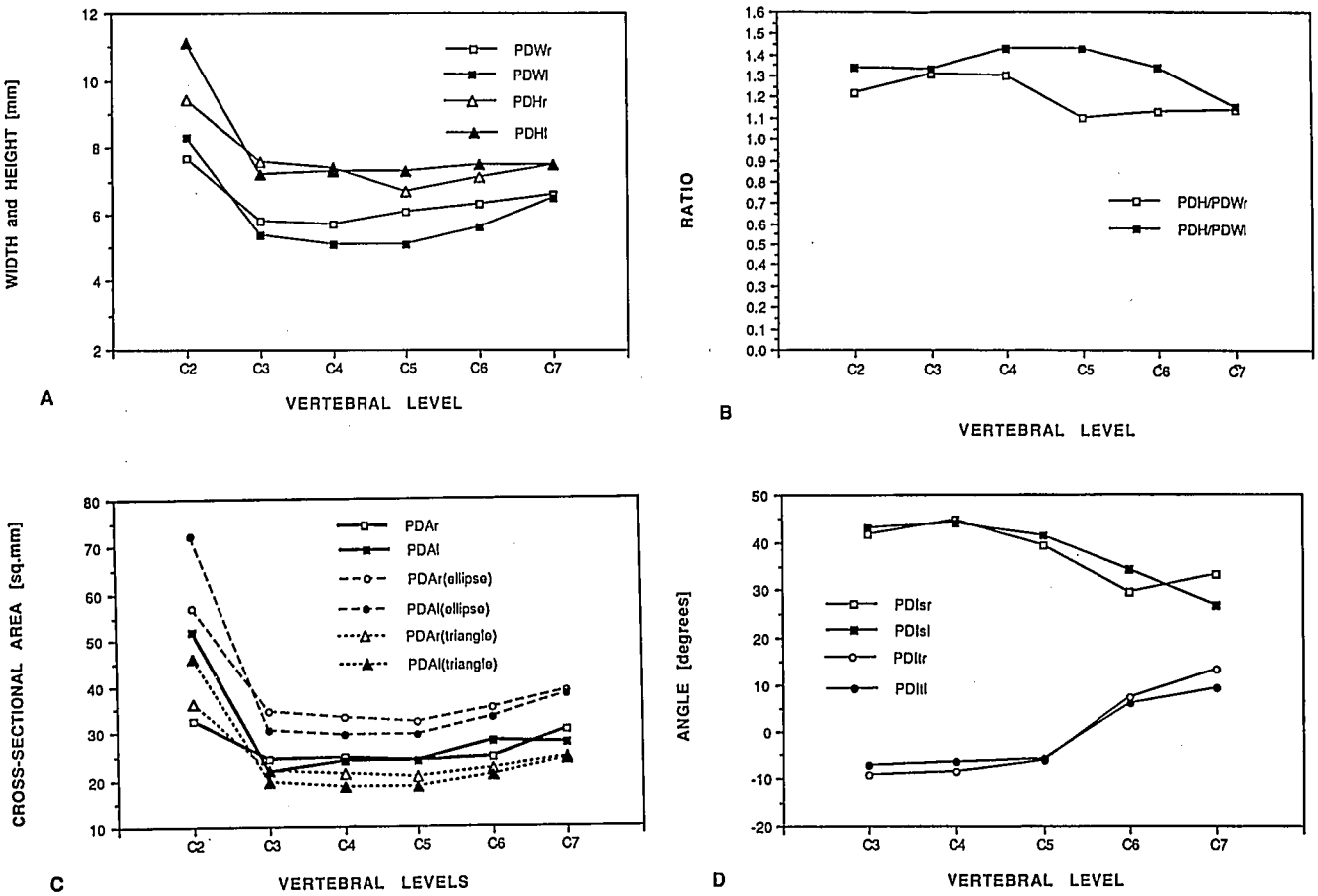



**Fig 4.** Pedicle dimensions as functions of vertebral levels C2–C7 are shown. **A,** Widths and heights of the pedicle for right and left. **B,** The ratio of height to width for the right and left pedicles. **C,** Cross-sectional areas of the left and right pedicles are shown by whole lines. The dash lines indicate the areas assuming these to be either an ellipse or a triangle. **D,** Pedicle axis angulations with respect to the sagittal and transverse planes for the right and left pedicles.

dimensions at C6 and C7 were significantly greater than those at the more superior vertebrae due to their greater load-bearing role. However at C7, the lower end-plate depth decreased, this change representing a distinct transition to the thoracic region. Another bony indicator of this transition was the spinal canal width-to-depth ratio, which jumped from an average cervical value of 1.4–1.6 at C7. Other indicators of the transition were the transverse process width and spinous process length at C7, which were much greater than the same at C6 and above.

Of particular interest to the transition between the cervical and thoracic regions were the dimensions of the spinal canal. The results of this study showed that the spinal canal area decreased from C6 to C7 in this region. Since the spinal cord remains approximately the same size

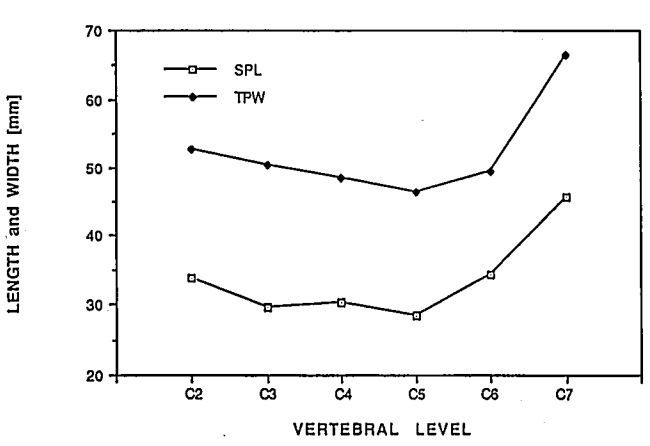



**Fig 5.** Spinous and transverse process dimensions as functions of the vertebral levels C2–C7.

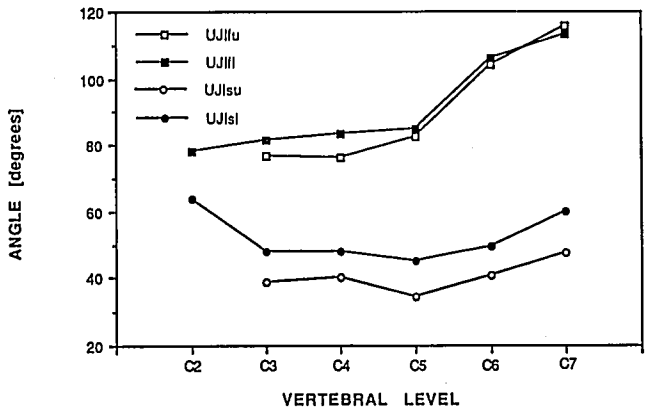



**Fig 6.** Uncovertebral joint angulations with respect to the frontal and sagittal planes for the upper and lower articulations are shown as functions of the vertebral levels C2–C7.

in this vicinity, a smaller spinal canal around the cord would create a greater potential for mechanical impingement of the cord during trauma. A high incidence of spinal cord injury at C6 has been observed by Fife and Kraus.[4]

Another obvious transition region represented in this study was from the upper cervical to the lower cervical from C2 to C3. The structures of the upper cervical region (C0–C1–C2) are unique because they support a large mass (head) in a region of high flexibility and highly vulnerable central nervous structures. The base of this joint complex is the C2 vertebra, and it has several features in its lower elements that differentiate it from the more inferior cervical vertebrae. The pedicle dimensions of C2 were much larger than those of vertebrae C3 through C7, and the spinal canal depth is much larger at the level of C2 than at lower levels. The large spinal canal depth provides sufficient mobility to the joint without encroaching on the spinal cord in this region. Since the spinal canal is very deep, the load-bearing facets are located in a more posterior direction. This requires the pedicles to have a greater cross-sectional area in bridging the gap between the facets and the vertebral body. In addition, there are several muscles that attach at the large spinous process of C2, thereby introducing large forces across the lamina and pedicles and requiring their sizes to be greater than at lower levels.

## Previous Studies

Some of the results of this study are compared with those of the earlier studies of Liu et al,[9] Nissan and Gilad[10] and Francis[5] (Table 7). In general, the results presented in this study agree well with those of the earlier studies, but there are some marked differences, as noted. The end-plate anteroposterior dimensions were within 2 mm of the data of Nissan and Gilad[10] and Francis[5] and followed the same trend from C2 through C7. The anteroposterior dimensions from Liu were somewhat smaller because their data were probably measured at the centrum midheight instead of the lower end-plate. The end-plate areas of our study differed significantly from those of Liu et al.[9] Our areas did not include the uncovertebral facet area; the values of Liu et al probably did. If one were to add our uncovertebral areas to the end-plate areas, the total areas would be very close to those of Francis.[5]

The spinal canal dimensions presented in this study were consistent with those of Francis. The only difference was that Francis did not observe widening of the canal at levels C5 and C6 and narrowing at C7 but saw a relatively small canal throughout.

The vertebral body heights presented herein were less than those observed by Nissan and Gilad[10] and Francis[5] by 2–2.5 mm. We believe this apparent discrepancy is due to the difference in where the height was measured. In our study, the height was the posterior body height in the midsagittal plane. In the Nissan and Gilad[10] and Francis[5] studies, the height measured was the posterior wall height as observed from the lateral direction. Due to the uncovertebral joints, the lateral projection of the height appears greater than it is in the midsagittal plane.

## Geometric Shape Approximations

To describe the shape of vertebral bodies in mathematical models, it is advantageous to represent the individual components of the vertebrae as simple geometric shapes. In this study, an attempt was made to model the areas of end plates, spinal canal, and pedicles as ellipses and triangles based on their measured widths and depths or heights.

For both the upper and lower end-plates (Figure 2C), the elliptical approximation was an overestimate of the measured area by 10% at all vertebral levels. The triangular approximation of both upper and lower end-plate shape was an underestimate by approximately 40%. Therefore, an elliptical approximation of cervical end plates appears justified.

For the spinal canal (Figure 3C), the elliptical approximation was better for levels C2 and C3, whereas the triangular approximation was better at levels C4 through C7. In all cases, the ellipse was an overestimate and the triangle an underestimate.

Table 7. Comparison with Previous Studies

| Parameter | Level | Present Study | Liu (1986) | Nissan (1984) | Francis (1952) |
|---|---|---|---|---|---|
| EPDI | C2 | 15.6 | — | 15.3 | 16.1 |
| | C3 | 15.6 | 14.4 | 15.6 | 16.4 |
| | C4 | 15.9 | 14.1 | 15.8 | 16.5 |
| | C5 | 17.9 | 14.8 | 16.1 | 16.8 |
| | C6 | 18.5 | 14.4 | 16.6 | 17.3 |
| | C7 | 16.8 | 14.5 | 16.3 | 16.7 |
| EPWI | C2 | 17.5 | — | — | 19.5 |
| | C3 | 17.2 | — | — | 20.5 |
| | C4 | 17.0 | — | — | 21.5 |
| | C5 | 19.4 | — | — | 22.5 |
| | C6 | 22.0 | — | — | 24.8 |
| | C7 | 23.4 | — | — | 29.3 |
| EPAI | C2 | 194.4 | 326.8 | — | 246.6 |
| | C3 | 190.7 | 309.7 | — | 264.0 |
| | C4 | 199.2 | 329.5 | — | 278.6 |
| | C5 | 246.2 | 361.1 | — | 296.9 |
| | C6 | 289.9 | 421.7 | — | 337.0 |
| | C7 | 280.3 | 440.5 | — | 384.3 |
| SCD | C2 | 21.0 | — | — | 22.0 |
| | C3 | 16.2 | — | — | 16.5 |
| | C4 | 17.7 | — | — | 15.4 |
| | C5 | 17.4 | — | — | 15.4 |
| | C6 | 18.1 | — | — | 15.4 |
| | C7 | 15.2 | — | — | 15.5 |
| SCW | C2 | 24.5 | — | — | 24.5 |
| | C3 | 22.9 | — | — | 23.9 |
| | C4 | 24.7 | — | — | 24.7 |
| | C5 | 24.9 | — | — | 25.6 |
| | C6 | 25.8 | — | — | 25.9 |
| | C7 | 24.5 | — | — | 25.6 |
| VBH | C3 | 11.6 | — | 14.5 | 14.3 |
| | C4 | 11.4 | — | 13.9 | 13.8 |
| | C5 | 11.4 | — | 13.8 | 13.3 |
| | C6 | 10.9 | — | 13.9 | 13.0 |
| | C7 | 12.8 | — | 14.9 | 14.6 |

*Note:* All dimensions are linear and their average values (mm) are given.

For the pedicles (Figure 4C), the elliptical approximation was larger than the measured areas, whereas the triangular approximation provided a good estimate of the pedicular area.

In summary, a comprehensive set of dimensions quantifying the three-dimensional vertebral anatomy of middle and lower cervical regions has been presented for the first time. The data describes average values and standard error of the means for some clinically important dimensions for each level and also provides variation from left to right.

We hope this information will improve the quality of mathematical models of the cervical spine by its incorporation of realistic vertebral anatomy. The data may also be used clinically, for example, in the design of surgical implants and procedures as well as for spinal instability determinations for this region of the spine.

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